

HealthConnect Clinic Experience Lab

Cross-Track Integration & Operational Findings Hand-Off

Hand-Off Context: Synthesizing analytical findings (\$N = 5,000\$, 48.46% no-show baseline, 29.64-day mean lead time) into concrete operational workstreams for Data Science, Engineering, and Operations pods.

Cross-Track Integration & Findings Hand-Off

Target Track	Key Findings Delivered	Practical Integration & Action
Data Science & ML Track	• Lead Time Predictor: Long booking windows (Mean = 29.64 days) serve as the primary risk feature. • Historical Flag: Previous no-show counts heavily weight patient drop-off probability.	• Feed these validated features directly into predictive classification models (e.g., Random Forest or XGBoost risk-scoring pipelines) to auto-flag high-risk patient profiles upon booking.
Engineering & Product Track	• Communication Efficacy: SMS dominance over email/WhatsApp for engagement. • Data Integrity: 100% Primary Key uniqueness and zero duplicates.	• Integrate an automated SMS gateway (e.g., Twilio / Africa's Talking) triggered at specific intervals (7, 3, and 1 days out). • Maintain strict database constraints enforced by initial audit logs.
Operations & Management Track	• Capacity Waste: 48.46% no-show rate vs. 46.28% attendance rate. • Logistical Friction: 10.11 km mean distance paired with 24.19-min wait times.	• Implement risk-based overbooking formulas for long-lead slots. • Introduce fast-track check-in queues for patients traveling > 10 km to reduce on-site attrition.

The Cross-Track Feedback Loop

To finalize project pod requirements, ensure documentation highlights the standard **Find** → **Act** → **Retest** → **Validate** cycle:

1. Find Analytics tracks identify the core operational bottleneck (e.g., lead-time decay).	2. Act Engineering/Operations deploy the intervention (e.g., automated SMS sequences).	3. Retest Data tracks pull post-intervention logs to measure changes in no-show rates.	4. Validate Final dashboards (app.py) reflect optimized KPIs for executive sign-off.
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